

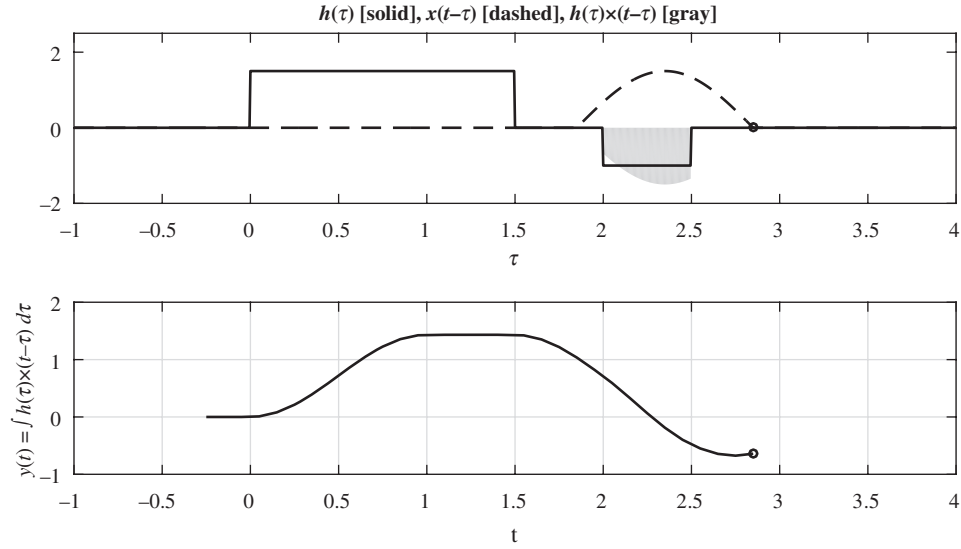


Figure 2.29 Graphical convolution at step $t = 2.85$ seconds.

2.8 APPENDIX: DETERMINING THE IMPULSE RESPONSE

In Eq. (2.13), we showed that for an LTIC system S specified by Eq. (2.11), the unit impulse response $h(t)$ can be expressed as

$$h(t) = b_0\delta(t) + \text{characteristic modes} \quad (2.51)$$

To determine the characteristic mode terms in Eq. (2.51), let us consider a system S_0 whose input $x(t)$ and the corresponding output $w(t)$ are related by

$$Q(D)w(t) = x(t) \quad (2.52)$$

Observe that both the systems S and S_0 have the same characteristic polynomial; namely, $Q(\lambda)$, and, consequently, the same characteristic modes. Moreover, S_0 is the same as S with $P(D) = 1$, that is, $b_0 = 0$. Therefore, according to Eq. (2.51), the impulse response of S_0 consists of characteristic mode terms only without an impulse at $t = 0$. Let us denote this impulse response of S_0 by $y_n(t)$. Observe that $y_n(t)$ consists of characteristic modes of S and therefore may be viewed as a zero-input response of S . Now $y_n(t)$ is the response of S_0 to input $\delta(t)$. Therefore, according to Eq. (2.52),

$$Q(D)y_n(t) = \delta(t)$$

or

$$(D^N + a_1D^{N-1} + \cdots + a_{N-1}D + a_N)y_n(t) = \delta(t)$$

or

$$y_n^{(N)}(t) + a_1y_n^{(N-1)}(t) + \cdots + a_{N-1}y_n^{(1)}(t) + a_Ny_n(t) = \delta(t)$$

where $y_n^{(k)}(t)$ represents the k th derivative of $y_n(t)$. The right-hand side contains a single impulse term, $\delta(t)$. This is possible only if $y_n^{(N-1)}(t)$ has a unit jump discontinuity at $t = 0$, so that $y_n^{(N)}(t) = \delta(t)$. Moreover, the lower-order terms cannot have any jump discontinuity because this would mean the presence of the derivatives of $\delta(t)$. Therefore $y_n(0) = y_n^{(1)}(0) = \dots = y_n^{(N-2)}(0) = 0$ (no discontinuity at $t = 0$), and the N initial conditions on $y_n(t)$ are

$$y_n(0) = y_n^{(1)}(0) = \dots = y_n^{(N-2)}(0) = 0 \quad \text{and} \quad y_n^{(N-1)}(0) = 1 \quad (2.53)$$

This discussion means that $y_n(t)$ is the zero-input response of the system S subject to initial conditions [Eq. (2.53)].

We now show that for the same input $x(t)$ to both systems, S and S_0 , their respective outputs $y(t)$ and $w(t)$ are related by

$$y(t) = P(D)w(t) \quad (2.54)$$

To prove this result, we operate on both sides of Eq. (2.52) by $P(D)$ to obtain

$$Q(D)P(D)w(t) = P(D)x(t)$$

Comparison of this equation with Eq. (2.2) leads immediately to Eq. (2.54).

Now if the input $x(t) = \delta(t)$, the output of S_0 is $y_n(t)$, and the output of S , according to Eq. (2.54), is $P(D)y_n(t)$. This output is $h(t)$, the unit impulse response of S . Note, however, that because it is an impulse response of a causal system S_0 , the function $y_n(t)$ is causal. To incorporate this fact we must represent this function as $y_n(t)u(t)$. Now it follows that $h(t)$, the unit impulse response of the system S , is given by

$$h(t) = P(D)[y_n(t)u(t)] \quad (2.55)$$

where $y_n(t)$ is a linear combination of the characteristic modes of the system subject to initial conditions (2.53).

The right-hand side of Eq. (2.55) is a linear combination of the derivatives of $y_n(t)u(t)$. Evaluating these derivatives is clumsy and inconvenient because of the presence of $u(t)$. The derivatives will generate an impulse and its derivatives at the origin. Fortunately when $M \leq N$ [Eq. (2.11)], we can avoid this difficulty by using the observation in Eq. (2.51), which asserts that at $t = 0$ (the origin), $h(t) = b_0\delta(t)$. Therefore, we need not bother to find $h(t)$ at the origin. This simplification means that instead of deriving $P(D)[y_n(t)u(t)]$, we can derive $P(D)y_n(t)$ and add to it the term $b_0\delta(t)$ so that

$$\begin{aligned} h(t) &= b_0\delta(t) + P(D)y_n(t) \quad t \geq 0 \\ &= b_0\delta(t) + [P(D)y_n(t)]u(t) \end{aligned}$$

This expression is valid when $M \leq N$ [the form given in Eq. (2.11)]. When $M > N$, Eq. (2.55) should be used.

2.9 SUMMARY

This chapter discusses time-domain analysis of LTIC systems. The total response of a linear system is a sum of the zero-input response and zero-state response. The zero-input response is the system

response generated only by the internal conditions (initial conditions) of the system, assuming that the external input is zero; hence the adjective “zero-input.” The zero-state response is the system response generated by the external input, assuming that all initial conditions are zero, that is, when the system is in zero state.

Every system can sustain certain forms of response on its own with no external input (zero input). These forms are intrinsic characteristics of the system; that is, they do not depend on any external input. For this reason they are called characteristic modes of the system. Needless to say, the zero-input response is made up of characteristic modes chosen in a combination required to satisfy the initial conditions of the system. For an N th-order system, there are N distinct modes.

The unit impulse function is an idealized mathematical model of a signal that cannot be generated in practice.[†] Nevertheless, introduction of such a signal as an intermediary is very helpful in analysis of signals and systems. The unit impulse response of a system is a combination of the characteristic modes of the system[‡] because the impulse $\delta(t) = 0$ for $t > 0$. Therefore, the system response for $t > 0$ must necessarily be a zero-input response, which, as seen earlier, is a combination of characteristic modes.

The zero-state response (response due to external input) of a linear system can be obtained by breaking the input into simpler components and then adding the responses to all the components. In this chapter we represent an arbitrary input $x(t)$ as a sum of narrow rectangular pulses [staircase approximation of $x(t)$]. In the limit as the pulse width $\rightarrow 0$, the rectangular pulse components approach impulses. Knowing the impulse response of the system, we can find the system response to all the impulse components and add them to yield the system response to the input $x(t)$. The sum of the responses to the impulse components is in the form of an integral, known as the convolution integral. The system response is obtained as the convolution of the input $x(t)$ with the system's impulse response $h(t)$. Therefore, the knowledge of the system's impulse response allows us to determine the system response to any arbitrary input.

LTIC systems have a very special relationship to the everlasting exponential signal e^{st} because the response of an LTIC system to such an input signal is the same signal within a multiplicative constant. The response of an LTIC system to the everlasting exponential input e^{st} is $H(s)e^{st}$, where $H(s)$ is the transfer function of the system.

If every bounded input results in a bounded output, the system is stable in the bounded-input/bounded-output (BIBO) sense. An LTIC system is BIBO-stable if and only if its impulse response is absolutely integrable. Otherwise, it is BIBO-unstable. BIBO stability is a stability seen from external terminals of the system. Hence, it is also called external stability or zero-state stability.

In contrast, internal stability (or the zero-input stability) examines the system stability from inside. When some initial conditions are applied to a system in zero state, then, if the system eventually returns to zero state, the system is said to be stable in the asymptotic or Lyapunov sense. If the system's response increases without bound, it is unstable. If the system does not go to zero state and the response does not increase indefinitely, the system is marginally stable. The internal stability criterion, in terms of the location of a system's characteristic roots, can be summarized as follows:

[†] However, it can be closely approximated by a narrow pulse of unit area and having a width that is much smaller than the time constant of an LTIC system in which it is used.

[‡] There is the possibility of an impulse in addition to the characteristic modes.

1. An LTIC system is asymptotically stable if, and only if, all the characteristic roots are in the LHP. The roots may be repeated or unrepeated.
2. An LTIC system is unstable if, and only if, either one or both of the following conditions exist: (i) at least one root is in the RHP; (ii) there are repeated roots on the imaginary axis.
3. An LTIC system is marginally stable if, and only if, there are no roots in the RHP, and there are some unrepeated roots on the imaginary axis.

It is possible for a system to be externally (BIBO) stable but internally unstable. When a system is controllable and observable, its external and internal descriptions are equivalent. Hence, external (BIBO) and internal (asymptotic) stabilities are equivalent and provide the same information. Such a BIBO-stable system is also asymptotically stable, and vice versa. Similarly, a BIBO-unstable system is either marginally stable or asymptotically unstable system.

The characteristic behavior of a system is extremely important because it determines not only the system response to internal conditions (zero-input behavior), but also the system response to external inputs (zero-state behavior) and the system stability. The system response to external inputs is determined by the impulse response, which itself is made up of characteristic modes. The width of the impulse response is called the time constant of the system, which indicates how fast the system can respond to an input. The time constant plays an important role in determining such diverse system behaviors as the response time and filtering properties of the system, dispersion of pulses, and the rate of pulse transmission through the system.

REFERENCES

1. Lathi, B. P., *Signals and Systems*. Berkeley-Cambridge Press, Carmichael, CA, 1987.
2. Mason, S. J., *Electronic Circuits, Signals, and Systems*. Wiley, New York, 1960.
3. Kailath, T., *Linear System*. Prentice-Hall, Englewood Cliffs, NJ, 1980.
4. Lathi, B. P., *Modern Digital and Analog Communication Systems*, 3rd ed. Oxford University Press, New York, 1998.

PROBLEMS

- 2.2-1** Determine the constants c_1 , c_2 , λ_1 , and λ_2 for each of the following second-order systems, which have zero-input responses of the form $y_{\text{zir}}(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$.
- (a) $\ddot{y}(t) + 2\dot{y}(t) + 5y(t) = \ddot{x}(t) - 5x(t)$ with $y_{\text{zir}}(0) = 2$ and $\dot{y}_{\text{zir}}(0) = 0$.
 - (b) $\ddot{y}(t) + 2\dot{y}(t) + 5y(t) = \ddot{x}(t) - 5x(t)$ with $y_{\text{zir}}(0) = 4$ and $\dot{y}_{\text{zir}}(0) = -1$.
 - (c) $\frac{d^2}{dt^2}y(t) + 2\frac{d}{dt}y(t) = x(t)$ with $y_{\text{zir}}(0) = 1$ and $\dot{y}_{\text{zir}}(0) = 2$.
 - (d) $(D^2 + 2D + 10)\{y(t)\} = (D^5 - D)\{x(t)\}$ with $y_{\text{zir}}(0) = \dot{y}_{\text{zir}}(0) = 1$.
 - (e) $(D^2 + \frac{7}{2}D + \frac{3}{2})\{y(t)\} = (D + 2)\{x(t)\}$ with $y_{\text{zir}}(0) = 3$ and $\ddot{y}_{\text{zir}}(0) = -8$. [Caution: The

second IC is given in terms of the second derivative, not the first derivative].

- (f) $13y(t) + 4\frac{d}{dt}y(t) + \frac{d^2}{dt^2}y(t) = 2x(t) - 4\frac{d}{dt}x(t)$ with $y_{\text{zir}}(0) = 3$ and $\dot{y}_{\text{zir}}(0) = -15$. [Caution: The second IC is given in terms of the second derivative, not the first derivative].

- 2.2-2** Consider a linear time-invariant system with input $x(t)$ and output $y(t)$ that is described by the differential equation

$$(D + 1)(D^2 - 1)\{y(t)\} = (D^5 - 1)\{x(t)\}$$

Furthermore, assume $y(0) = \dot{y}(0) = \ddot{y}(0) = 1$.

- (a) What is the order of this system?
 (b) What are the characteristic roots of this system?
 (c) Determine the zero-input response $y_{\text{zir}}(t)$. Simplify your answer.

2.2-3 A real LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant-coefficient linear differential equation:

$$(D^3 + 9D)\{y(t)\} = (2D^3 + 1)\{x(t)\}.$$

- (a) What is the characteristic equation of this system?
 (b) What are the characteristic modes of this system?
 (c) Assuming $y_{\text{zir}}(0) = 4$, $\dot{y}_{\text{zir}}(0) = -18$, and $\ddot{y}_{\text{zir}}(0) = 0$, determine this system's zero-input response $y_{\text{zir}}(t)$. Simplify $y_{\text{zir}}(t)$ to include only real terms (i.e., no j 's should appear in your answer).

2.2-4 An LTIC system is specified by the equation

$$(D^2 + 5D + 6)y(t) = (D + 1)x(t)$$

- (a) Find the characteristic polynomial, characteristic equation, characteristic roots, and characteristic modes of this system.
 (b) Find $y_0(t)$, the zero-input component of the response $y(t)$ for $t \geq 0$, if the initial conditions are $y_0(0^-) = 2$ and $\dot{y}_0(0^-) = -1$.

2.2-5 Repeat Prob. 2.2-4 for

$$(D^2 + 4D + 4)y(t) = Dx(t)$$

and $y_0(0^-) = 3$, $\dot{y}_0(0^-) = -4$.

2.2-6 Repeat Prob. 2.2-4 for

$$D(D + 1)y(t) = (D + 2)x(t)$$

and $y_0(0^-) = \dot{y}_0(0^-) = 1$.

2.2-7 Repeat Prob. 2.2-4 for

$$(D^2 + 9)y(t) = (3D + 2)x(t)$$

and $y_0(0^-) = 0$, $\dot{y}_0(0^-) = 6$.

2.2-8 Repeat Prob. 2.2-4 for

$$(D^2 + 4D + 13)y(t) = 4(D + 2)x(t)$$

with $y_0(0^-) = 5$, $\dot{y}_0(0^-) = 15.98$.

2.2-9 Repeat Prob. 2.2-4 for

$$D^2(D + 1)y(t) = (D^2 + 2)x(t)$$

with $y_0(0^-) = 4$, $\dot{y}_0(0^-) = 3$, and $\ddot{y}_0(0^-) = -1$.

2.2-10 Repeat Prob. 2.2-4 for

$$(D + 1)(D^2 + 5D + 6)y(t) = Dx(t)$$

with $y_0(0^-) = 2$, $\dot{y}_0(0^-) = -1$, and $\ddot{y}_0(0^-) = 5$.

2.2-11 A system is described by a constant-coefficient linear differential equation and has zero-input response given by $y_0(t) = 2e^{-t} + 3$.

- (a) Is it possible for the system's characteristic equation to be $\lambda + 1 = 0$? Justify your answer.
 (b) Is it possible for the system's characteristic equation to be $\sqrt{3}(\lambda^2 + \lambda) = 0$? Justify your answer.
 (c) Is it possible for the system's characteristic equation to be $\lambda(\lambda + 1)^2 = 0$? Justify your answer.

2.2-12 Consider the circuit of Fig. P2.2-12. Using operator notation, this system can be described as $(D + a_1)\{y(t)\} = (b_0D + b_1)\{x(t)\}$.

- (a) Determine the constants a_1 , b_0 , and b_1 in terms of the system components R , R_f , and C .
 (b) Assume that $R = 300 \text{ k}\Omega$, $R_f = 1.2 \text{ M}\Omega$, and $C = 5 \text{ }\mu\text{F}$. What is the zero-input response $y_0(t)$ of this system, assuming $v_C(0) = 1 \text{ V}$?

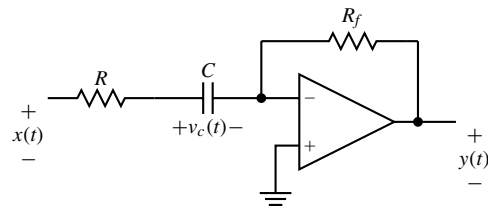


Figure P2.2-12

2.3-1 Determine the characteristic equation, characteristic modes, and impulse response $h(t)$ for each of the following real LTIC systems. Since the systems are real, express each $h(t)$ using only real terms (i.e., no j 's should appear in your answers).

- (a) $(D^2 + 1)\{y(t)\} = 2D\{x(t)\}$
 (b) $(D^3 + D)\{y(t)\} = (2D^3 + 1)\{x(t)\}$
 (c) $\frac{d^2}{dt^2}y(t) + 2\frac{d}{dt}y(t) + 5y(t) = 8x(t)$

2.3-2 Find the unit impulse response of a system specified by the equation

$$(D^2 + 4D + 3)y(t) = (D + 5)x(t)$$

2.3-3 Repeat Prob. 2.3-2 for

$$(D^2 + 5D + 6)y(t) = (D^2 + 7D + 11)x(t)$$

2.3-4 Repeat Prob. 2.3-2 for the first-order allpass filter specified by the equation

$$(D + 1)y(t) = -(D - 1)x(t)$$

2.3-5 Find the unit impulse response of an LTIC system specified by the equation

$$(D^2 + 6D + 9)y(t) = (2D + 9)x(t)$$

2.3-6 Determine and plot the unit impulse response $h(t)$ of the op-amp circuit of Fig. P2.2-12, assuming that $R = 300 \text{ k}\Omega$, $R_f = 1.2 \text{ M}\Omega$, and $C = 5 \text{ }\mu\text{F}$.

2.3-7 A causal LTIC system with input $x(t)$ and output $y(t)$ is described by the constant coefficient integro-differential equation

$$y(t) + \int 3y(t) dt + \iint 2y(t) dt = \iint x(t) dt - \iiint x(t) dt.$$

- (a) Express this system as a constant coefficient linear differential equation in standard operator form.
 (b) Determine the characteristic modes of this system.

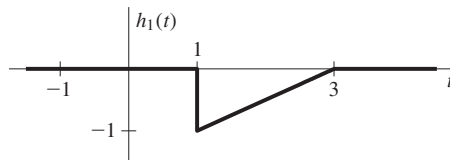


Figure P2.4-1

- (c) Determine the impulse response $h(t)$ of this system.

2.4-1 Let $f(t) = h_1(t) * h_2(t)$, where $h_1(t)$ and $h_2(t)$ are shown in Fig. P2.4-1. In the following, use the graphical convolution procedure where you flip and shift $h_2(t)$.

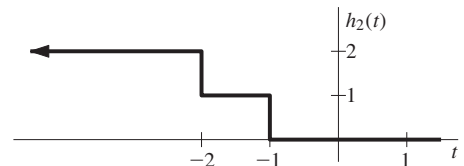
- (a) Plot $h_1(\tau)$ and $h_2(t - \tau)$ as functions of τ . Clearly label the plots, including necessary function parameterizations.
 (b) Determine the (piecewise) regions of $f(t)$ and set up the corresponding integrals that describe $f(t)$ in those regions. Do not evaluate the integrals, only set them up!
 (c) Determine $f(1)$, which is $f(t)$ evaluated at $t = 1$. Provide a number, not a formula.

2.4-2 Consider signals $h(t) = u(t + 3) - 2u(t + 1) + u(t - 1)$ and $x(t) = \cos(t) [u(t - \pi/2) - u(t - 3\pi/2)]$. Let $y(t) = x(t) * h(t)$.

- (a) Determine the last time t_{last} that $y(t)$ is nonzero. That is, find the smallest value t_{last} such that $y(t) = 0$ for all $t > t_{\text{last}}$.
 (b) Determine the approximate time t_{max} where $y(t)$ is a maximum.

2.4-3 Consider signals $h(t) = -u(t + 2) + 3u(t - 1) - 2u(t - \frac{5}{2})$ and $x(t) = \sin(t) [u(t + 2\pi) - u(t + \pi)]$. Determine the approximate time t_{min} where $y(t) = x(t) * h(t)$ is a minimum. Note, the minimum value of $y(t) \neq 0$!

2.4-4 An LTIC system has impulse response $h(t) = 3u(t - 2)$. For input $x(t)$ shown in Fig. P2.4-4, use the graphical convolution procedure to determine $y_{\text{zsr}}(t) = h(t) * x(t)$. Accurately sketch $y_{\text{zsr}}(t)$. When solving for $y_{\text{zsr}}(t)$, flip and shift $x(t)$ and explicitly show all integration steps—even if apparently trivial!



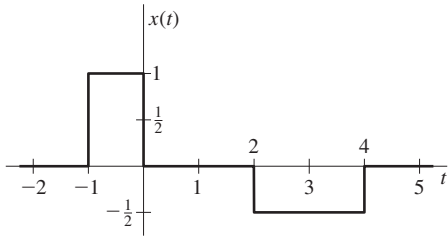


Figure P2.4-4

- 2.4-5** Suppose an LTIC system has impulse response $h(t)$ and input $x(t) = u(t)$. Figure P2.4-5 shows $x(t)$ and $h(t+1)$, respectively. Be careful! Figure P2.4-5 shows $h(t+1)$, not $h(t)$.

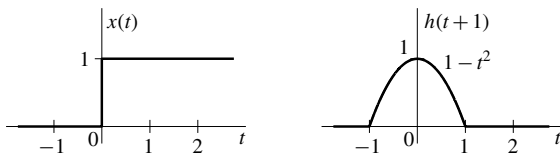


Figure P2.4-5

- Is system $h(t)$ causal? Mathematically justify your answer.
- Use the graphical convolution procedure to determine $y_{\text{zsr}}(t) = x(t) * h(t)$. Accurately sketch $y_{\text{zsr}}(t)$. When solving for $y_{\text{zsr}}(t)$, flip and shift $x(t)$ and explicitly show all integration steps.

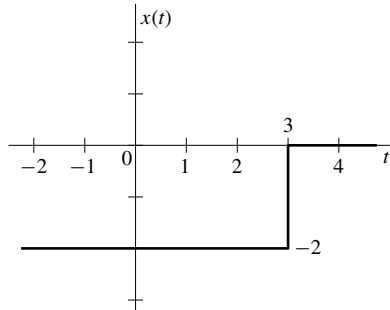


Figure P2.4-7

- 2.4-6** Repeat Prob. 2.4-5 using the signals of Fig. P2.4-6, rather than those of Fig. P2.4-5. Be careful! Figure P2.4-6 shows $h(t-1)$, not $h(t)$.

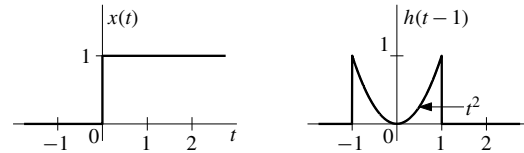
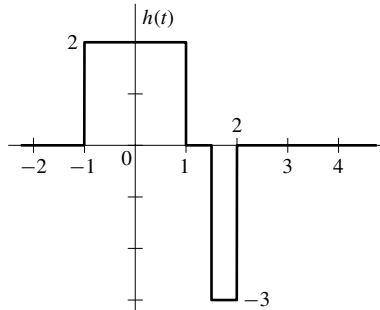


Figure P2.4-6

- 2.4-7** Suppose an LTIC system has impulse response $h(t)$ and input $x(t)$, both shown in Fig. P2.4-7. Use the graphical convolution procedure to determine $y_{\text{zsr}}(t) = x(t) * h(t)$. Accurately sketch $y_{\text{zsr}}(t)$. When solving for $y_{\text{zsr}}(t)$, flip and shift $x(t)$ and explicitly show all integration steps.
- 2.4-8** An LTIC system has impulse response $h(t)$, as shown in Fig. P2.4-8. Let t have units of seconds. Let the input be $x(t) = u(-t-2)$ and designate the output as $y_{\text{zsr}}(t) = x(t) * h(t)$.
- Use the graphical convolution procedure where $h(t)$ is flipped and shifted to determine $y_{\text{zsr}}(t)$. Accurately plot your result.
 - Use the graphical convolution procedure where $x(t)$ is flipped and shifted to determine $y_{\text{zsr}}(t)$. Accurately plot your result.
- 2.4-9** If $c(t) = x(t) * g(t)$, then show that $A_c = A_x A_g$, where A_x, A_g , and A_c are the areas under $x(t)$, $g(t)$, and $c(t)$, respectively. Verify this *area property* of convolution in Exs. 2.10 and 2.12.



- 2.4-10** If $x(t) * g(t) = c(t)$, then show that $x(at) * g(at) = |1/a|c(at)$. This *time-scaling property* of convolution states that if both $x(t)$ and $g(t)$ are time-scaled by a , their convolution is also time-scaled by a (and multiplied by $|1/a|$).

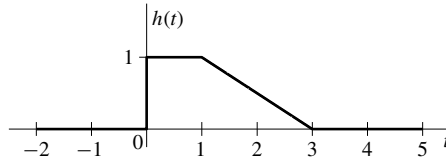


Figure P2.4-8

- 2.4-11** Show that the convolution of an odd and an even function is an odd function and the convolution of two odd or two even functions is an even function. [Hint: Use the time-scaling property of convolution in Prob. 2.4-10.]
- 2.4-12** Suppose an LTIC system has impulse response $h(t) = (1-t)[u(t) - u(t-1)]$ and input $x(t) = u(-t-1) + u(t-1)$. Use the graphical convolution procedure to determine $y_{\text{zsr}}(t) = x(t) * h(t)$. Accurately sketch $y_{\text{zsr}}(t)$. When solving for $y_{\text{zsr}}(t)$, flip and shift $h(t)$, explicitly show all integration steps, and simplify your answer.
- 2.4-13** Using direct integration, find $e^{-at}u(t) * e^{-bt}u(t)$.
- 2.4-14** Using direct integration, find $u(t) * u(t)$, $e^{-at}u(t) * e^{-at}u(t)$, and $tu(t) * u(t)$.
- 2.4-15** Using direct integration, find $\sin tu(t) * u(t)$ and $\cos tu(t) * u(t)$.
- 2.4-16** The unit impulse response of an LTIC system is

$$h(t) = e^{-t}u(t)$$

Find this system's (zero-state) response $y(t)$ if the input $x(t)$ is:

- (a) $u(t)$
- (b) $e^{-t}u(t)$
- (c) $e^{-2t}u(t)$
- (d) $\sin 3tu(t)$

Use the convolution table (Table 2.1) to find your answers.

- 2.4-17** Repeat Prob. 2.4-16 for

$$h(t) = [2e^{-3t} - e^{-2t}]u(t)$$

and if the input $x(t)$ is:

- (a) $u(t)$
- (b) $e^{-t}u(t)$
- (c) $e^{-2t}u(t)$

- 2.4-18** Repeat Prob. 2.4-16 for

$$h(t) = (1-2t)e^{-2t}u(t)$$

and input $x(t) = u(t)$.

- 2.4-19** Repeat Prob. 2.4-16 for

$$h(t) = 4e^{-2t} \cos 3tu(t)$$

and each of the following inputs $x(t)$:

- (a) $u(t)$
- (b) $e^{-t}u(t)$

- 2.4-20** Repeat Prob. 2.4-16 for

$$h(t) = e^{-t}u(t)$$

and each of the following inputs $x(t)$:

- (a) $e^{-2t}u(t)$
- (b) $e^{-2(t-3)}u(t)$
- (c) $e^{-2t}u(t-3)$
- (d) The gate pulse depicted in Fig. P2.4-20—and provide a sketch of $y(t)$.

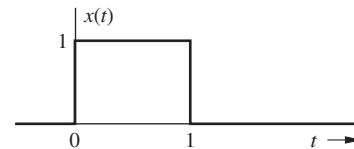


Figure P2.4-20

- 2.4-21** A first-order allpass filter impulse response is given by

$$h(t) = -\delta(t) + 2e^{-t}u(t)$$

- (a) Find the zero-state response of this filter for the input $e^t u(-t)$.
- (b) Sketch the input and the corresponding zero-state response.

- 2.4-22** Figure P2.4-22 shows the input $x(t)$ and the impulse response $h(t)$ for an LTIC system. Let the output be $y(t)$.

- (a) By inspection of $x(t)$ and $h(t)$, find $y(-1), y(0), y(1), y(2), y(3), y(4), y(5)$, and

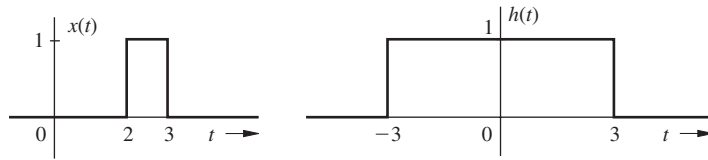


Figure P2.4-22

- $y(6)$. Thus, by merely examining $x(t)$ and $h(t)$, you are required to see what the result of convolution yields at $t = -1, 0, 1, 2, 3, 4, 5$, and 6 .
- (b) Find the system response to the input $x(t)$.
- 2.4-23** The zero-state response of an LTIC system to an input $x(t) = 2e^{-2t}u(t)$ is $y(t) = [4e^{-2t} + 6e^{-3t}]u(t)$. Find the impulse response of the system. [Hint: We have not yet developed a method of finding $h(t)$ from the knowledge of the input and the corresponding output. Knowing the form of $x(t)$ and $y(t)$, you will have to make the best guess of the general form of $h(t)$.]
- 2.4-24** Sketch the functions $x(t) = 1/(t^2 + 1)$ and $u(t)$. Now find $x(t) * u(t)$ and sketch the result.
- 2.4-25** Figure P2.4-25 shows $x(t)$ and $g(t)$. Find and sketch $c(t) = x(t) * g(t)$.
- 2.4-26** Find and sketch $c(t) = x(t) * g(t)$ for the functions depicted in Fig. P2.4-26.
- 2.4-27** Find and sketch $c(t) = x_1(t) * x_2(t)$ for the pairs of functions illustrated in Fig. P2.4-27.
- 2.4-28** Use Eq. (2.37) to find the convolution of $x(t)$ and $w(t)$, shown in Fig. P2.4-28.
- 2.4-29** Determine $H(s)$, the transfer function of an ideal time delay of T seconds. Find your answer by two methods: using Eq. (2.39) and using Eq. (2.40).
- 2.4-30** Determine $y(t) = x(t) * h(t)$ for the signals depicted in Fig. P2.4-30.
- 2.4-31** Two linear time-invariant systems, each with impulse response $h(t)$, are connected in cascade. Refer to Fig. P2.4-31. Given input $x(t) = u(t)$, determine $y(1)$. That is, determine the step response at time $t = 1$ for the cascaded system shown.
- 2.4-32** Consider the electric circuit shown in Fig. P2.4-32.
- Determine the differential equation that relates the input $x(t)$ to output $y(t)$. Recall that $i_C(t) = C \frac{dv_C(t)}{dt}$ and $v_L(t) = L \frac{di_L(t)}{dt}$.
 - Find the characteristic equation for this circuit, and express the root(s) of the characteristic equation in terms of L and C .
 - Determine the zero-input response given an initial capacitor voltage of one volt and an initial inductor current of zero amps. That is, find $y_0(t)$ given $v_C(0) = 1$ V and

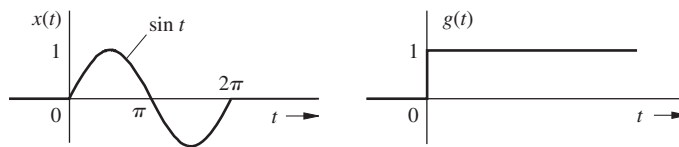


Figure P2.4-25

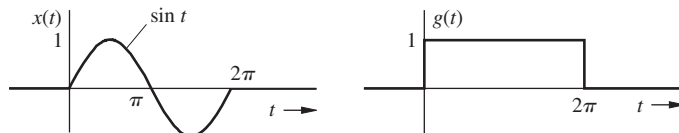


Figure P2.4-26

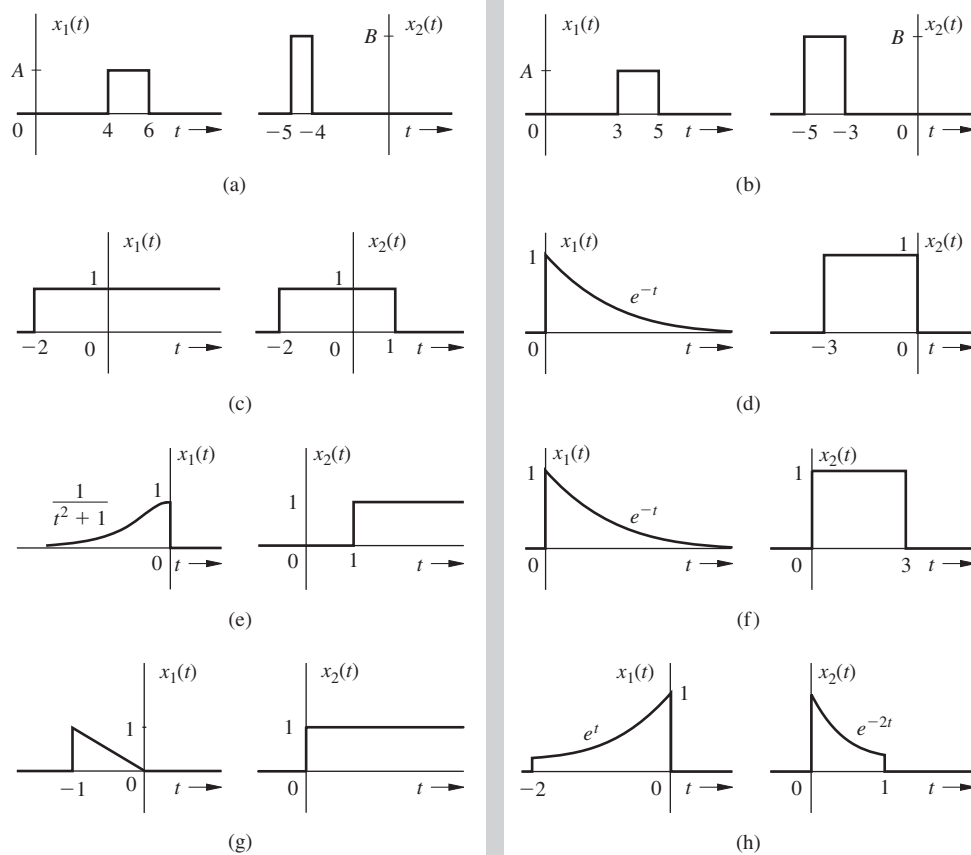


Figure P2.4-27

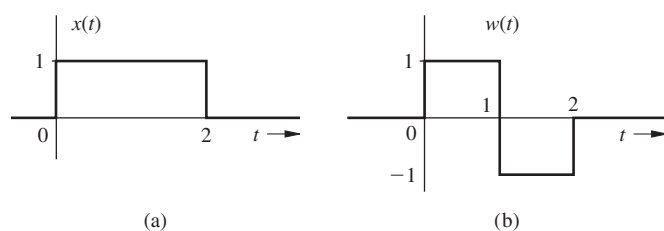


Figure P2.4-28

- $i_L(0) = 0$ A. [Hint: The coefficient(s) in $y_0(t)$ are independent of L and C .]
 (d) Plot $y_0(t)$ for $t \geq 0$. Does the zero-input response, which is caused solely by initial conditions, ever “die out”?

- (e) Determine the total response $y(t)$ to the input $x(t) = e^{-t}u(t)$. Assume an initial inductor current of $i_L(0^-) = 0$ A, an initial capacitor voltage of $v_C(0^-) = 1$ V, $L = 1$ H, and $C = 1$ F.

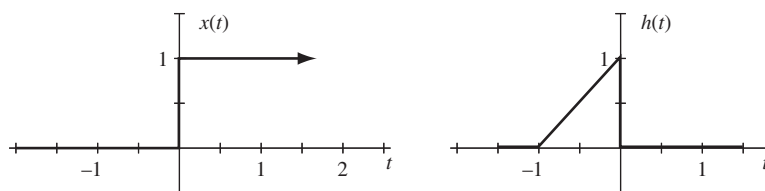


Figure P2.4-30

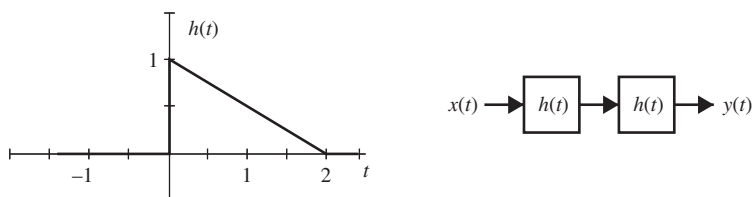


Figure P2.4-31

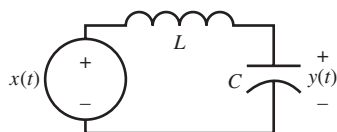


Figure P2.4-32

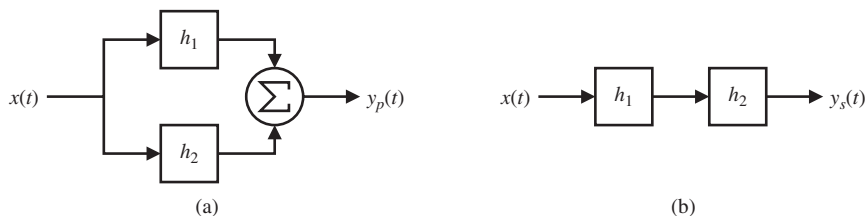


Figure P2.4-33

- 2.4-33** Two LTIC systems have impulse response functions given by $h_1(t) = (1-t)[u(t) - u(t-1)]$ and $h_2(t) = t[u(t+2) - u(t-2)]$.
- Carefully sketch the functions $h_1(t)$ and $h_2(t)$.
 - Assume that the two systems are connected in parallel, as shown in Fig. P2.4-33a. Carefully plot the equivalent impulse response function, $h_p(t)$.
 - Assume that the two systems are connected in cascade, as shown in Fig. P2.4-33b. Carefully plot the equivalent impulse response function, $h_s(t)$.

- 2.4-34** Consider the circuit shown in Fig. P2.4-34.
- Find the output $y(t)$ given an initial capacitor voltage of $y(0) = 2$ volts and an input $x(t) = u(t)$.
 - Given an input $x(t) = u(t-1)$, determine the initial capacitor voltage $y(0)$ so that the output $y(t)$ is 0.5 volt at $t = 2$ seconds.

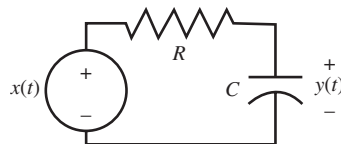


Figure P2.4-34

- 2.4-35** An analog signal is given by $x(t) = t[u(t) - u(t-1)]$, as shown in Fig. P2.4-35. Determine and plot $y(t) = x(t) * x(2t)$.

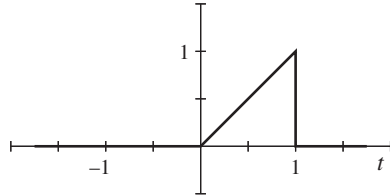


Figure P2.4-35

- 2.4-36** Consider the electric circuit shown in Fig. P2.4-36.
- Determine the differential equation that relates the input current $x(t)$ to output current $y(t)$. Recall that

$$v_L(t) = L \frac{di_L(t)}{dt}$$

- Find the characteristic equation for this circuit, and express the root(s) of the characteristic equation in terms of L_1 , L_2 , and R .
- Determine the zero-input response given initial inductor currents of one ampere each. That is, find $y_0(t)$ given $i_{L_1}(0) = i_{L_2}(0) = 1$ A.

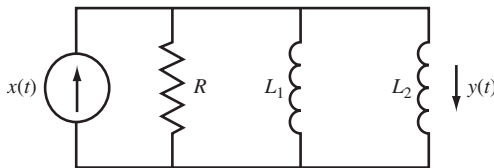


Figure P2.4-36

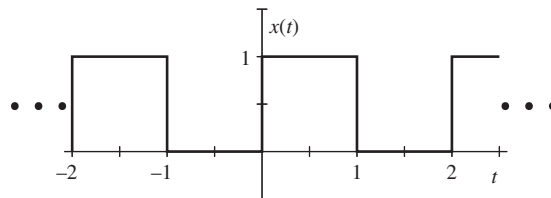


Figure P2.4-38

- 2.4-37** An LTI system has step response given by $g(t) = e^{-t}u(t) - e^{-2t}u(t)$. Determine the output of this system $y(t)$ given an input $x(t) = \delta(t - \pi) - \cos(\sqrt{3})u(t)$.

- 2.4-38** The periodic signal $x(t)$ shown in Fig. P2.4-38 is input to a system with impulse response function $h(t) = t[u(t) - u(t-1.5)]$, also shown in Fig. P2.4-38. Use convolution to determine the output $y(t)$ of this system. Plot $y(t)$ over $(-3 \leq t \leq 3)$.

- 2.4-39** Consider the electric circuit shown in Fig. P2.4-39.

- Determine the differential equation relating input $x(t)$ to output $y(t)$.
- Determine the output $y(t)$ in response to the input $x(t) = 4te^{-3t/2}u(t)$. Assume component values of $R = 1 \Omega$, $C_1 = 1$ F, and $C_2 = 2$ F, and initial capacitor voltages of $V_{C_1} = 2$ V and $V_{C_2} = 1$ V.

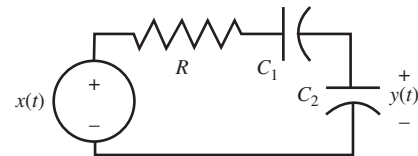
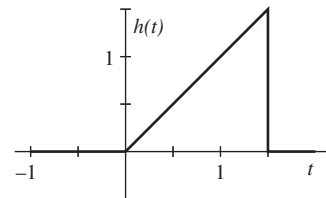


Figure P2.4-39

- 2.4-40** An LTIC system has impulse response $h(t) = 3e^{-|t|}$.

- Is the system causal? Mathematically justify your answer.
- Determine the zero-state response of this system if the input is $x(t) = u(2-t)$.

- 2.4-41** A cardiovascular researcher is attempting to model the human heart. He has recorded ventricular pressure, which he believes corresponds to the heart's impulse response



function $h(t)$, as shown in Fig. P2.4-41. Comment on the function $h(t)$ shown in Fig. P2.4-41. Can you establish any system properties, such as causality or stability? Do the data suggest any reason to suspect that the measurement is not a true impulse response?

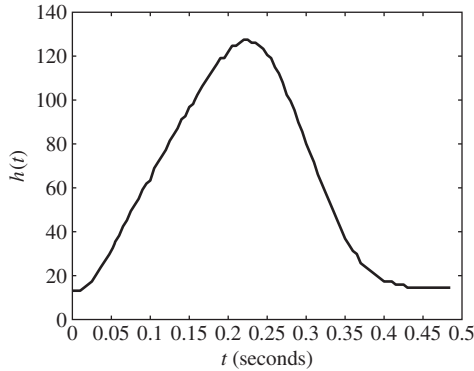


Figure P2.4-41

2.4-42 Consider an integrator system, $y(t) = \int_{-\infty}^t x(\tau) d\tau$.

- What is the unit impulse response $h_i(t)$ of this system?
- If two such integrators are put in parallel, what is the resulting impulse response $h_p(t)$?
- If two such integrators are put in series, what is the resulting impulse response $h_s(t)$?

2.4-43 The autocorrelation of a function $x(t)$ is given by $r_{xx}(t) = \int_{-\infty}^{\infty} x(\tau)x(\tau - t) d\tau$. This equation is computed in a manner nearly identical to convolution.

- Show $r_{xx}(t) = x(t) * x(-t)$.
- Determine and plot $r_{xx}(t)$ for the signal $x(t)$ depicted in Fig. P2.4-43. [Hint: $r_{xx}(t) = r_{xx}(-t)$.]

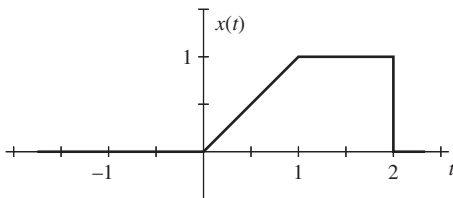


Figure P2.4-43

2.4-44 Consider the circuit shown in Fig. P2.4-44. This circuit functions as an integrator. Assume ideal op-amp behavior and recall that

$$i_C(t) = C \frac{dV_C(t)}{dt}$$

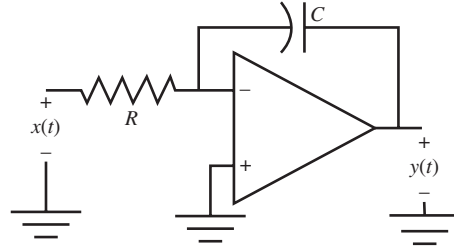


Figure P2.4-44

- Determine the differential equation that relates the input $x(t)$ to the output $y(t)$.
- This circuit does not behave well at dc. Demonstrate this by computing the zero-state response $y(t)$ for a unit step input $x(t) = u(t)$.

2.4-45 Derive the result in Eq. (2.37) in another way. As mentioned in Ch. 1 (Fig. 1.27b), it is possible to express an input in terms of its step components, as shown in Fig. P2.4-45. Find the system response as a sum of the responses to the step components of the input.

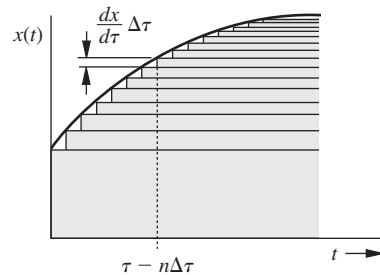


Figure P2.4-45

2.4-46 Show that an LTIC system response to an everlasting sinusoid $\cos \omega_0 t$ is given by

$$y(t) = |H(j\omega_0)| \cos[\omega_0 t + \angle H(j\omega_0)]$$

where

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt$$

assuming the integral on the right-hand side exists.

- 2.4-47** A line charge is located along the x axis with a charge density $Q(x)$ coulombs per meter. Show that the electric field $E(x)$ produced by this line charge at a point x is given by

$$E(x) = Q(x) * h(x)$$

where $h(x) = 1/4\pi\epsilon x^2$. [Hint: The charge over an interval $\Delta\tau$ located at $\tau = n\Delta\tau$ is $Q(n\Delta\tau)\Delta\tau$. Also by Coulomb's law, the electric field $E(r)$ at a distance r from a charge q coulombs is given by $E(r) = q/4\pi\epsilon r^2$.]

- 2.4-48** A system is called complex if a real-valued input can produce a complex-valued output. Suppose a linear time-invariant complex system has impulse response $h(t) = j[u(-t+2) - u(-t)]$.
- Is this system causal? Explain.
 - Use convolution to determine the zero-state response $y_1(t)$ of this system in response to the unit-duration pulse $x_1(t) = u(t) - u(t-1)$.
 - Using the result from part (a), determine the zero-state response $y_2(t)$ in response to $x_2(t) = 2u(t-1) - u(t-2) - u(t-3)$.
- 2.5-1** Explain, with reasons, whether the LTIC systems described by the following equations are (i) stable or unstable in the BIBO sense; (ii) asymptotically stable, unstable, or marginally stable. Assume that the systems are controllable and observable.
- $(D^2 + 8D + 12)y(t) = (D - 1)x(t)$
 - $D(D^2 + 3D + 2)y(t) = (D + 5)x(t)$
 - $D^2(D^2 + 2)y(t) = x(t)$
 - $(D + 1)(D^2 - 6D + 5)y(t) = (3D + 1)x(t)$
- 2.5-2** Repeat Prob. 2.5-1 for the following:
- $(D + 1)(D^2 + 2D + 5)^2y(t) = x(t)$
 - $(D + 1)(D^2 + 9)y(t) = (2D + 9)x(t)$
 - $(D + 1)(D^2 + 9)^2y(t) = (2D + 9)x(t)$
 - $(D^2 + 1)(D^2 + 4)(D^2 + 9)y(t) = 3Dx(t)$
- 2.5-3** Consider an LTIC system with unit impulse response $h(t) = e^t \left[\frac{2}{3} \cos(\frac{3}{2}t) + \frac{1}{3} \sin(\pi t) \right] u(123 - t)$. Is this system BIBO-stable? Mathematically justify your answer.

- 2.5-4** Consider an LTIC system with unit impulse response $h(t) = \frac{1}{t}u(t-T)$.
- Determine, if possible, the value(s) of T for which this system is causal.
 - Determine, if possible, the value(s) of T for which this system is BIBO-stable.
- Justify all answers mathematically.
- 2.5-5** You are given the choice of a system that is guaranteed internally stable or a system that is guaranteed externally stable. Which do you choose? Why?
- 2.5-6** For a certain LTIC system, the impulse response $h(t) = u(t)$.
- Determine the characteristic root(s) of this system.
 - Is this system asymptotically or marginally stable, or is it unstable?
 - Is this system BIBO-stable?
 - What can this system be used for?
- 2.5-7** In Sec. 2.5 we demonstrated that for an LTIC system, the condition of Eq. (2.45) is sufficient for BIBO stability. Show that this is also a necessary condition for BIBO stability in such systems. In other words, show that if Eq. (2.45) is not satisfied, then there exists a bounded input that produces an unbounded output. [Hint: Assume that a system exists for which $h(t)$ violates Eq. (2.45) and yet produces an output that is bounded for every bounded input. Establish the contradiction in this statement by considering an input $x(t)$ defined by $x(t_1 - \tau) = 1$ when $h(\tau) \geq 0$ and $x(t_1 - \tau) = -1$ when $h(\tau) < 0$, where t_1 is some fixed instant.]
- 2.5-8** An analog LTIC system with impulse response function $h(t) = u(t+2) - u(t-2)$ is presented with an input $x(t) = t(u(t) - u(t-2))$.
- Determine and plot the system output $y(t) = x(t) * h(t)$.
 - Is this system stable? Is this system causal? Justify your answers.
- 2.5-9** A system has an impulse response function shaped like a rectangular pulse, $h(t) = u(t) - u(t-1)$. Is the system stable? Is the system causal?
- 2.5-10** A continuous-time LTI system has impulse response function $h(t) = \sum_{i=0}^{\infty} (0.5)^i \delta(t-i)$.
- Is the system causal? Prove your answer.
 - Is the system stable? Prove your answer.

- 2.6-1** Data at a rate of 1 million pulses per second are to be transmitted over a certain communications channel. The unit step response $g(t)$ for this channel is shown in Fig. P2.6-1.

- (a) Can this channel transmit data at the required rate? Explain your answer.
 (b) Can an audio signal consisting of components with frequencies up to 15 kHz be transmitted over this channel with reasonable fidelity?

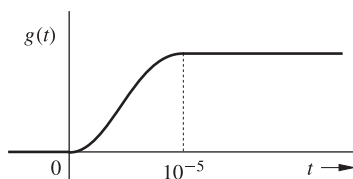


Figure P2.6-1

- 2.6-2** Determine a frequency ω that will cause the input $x(t) = \cos(\omega t)$ to produce a strong response when applied to the system described by $(D^2 + 2D + 13/4)\{y(t)\} = x(t)$. Carefully explain your choice.

- 2.6-3** Figure P2.6-3 shows the impulse response $h(t)$ of a lowpass LTIC system. Determine the peak amplitude A and time constant T_h so that rectangular impulse response $\hat{h}(t)$ is an appropriate approximation of $h(t)$. The two graphs of Fig. P2.6-3 are not necessarily drawn to the same scale.

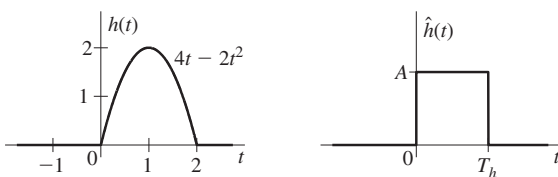


Figure P2.6-3

- 2.6-4** A certain communication channel has a bandwidth of 10 kHz. A pulse of 0.5 ms duration is transmitted over this channel.

- (a) Determine the width (duration) of the received pulse.
 (b) Find the maximum rate at which these pulses can be transmitted over this channel without interference between the successive pulses.

- 2.6-5** A first-order LTIC system has a characteristic root $\lambda = -10^4$.

- (a) Determine T_r , the rise time of its unit step input response.
 (b) Determine the bandwidth of this system.
 (c) Determine the rate at which the information pulses can be transmitted through this system.

- 2.6-6** A lowpass system with a 6 MHz cutoff frequency needs to transmit data pulse that are $\frac{500}{6}$ ns wide. Determine a suitable transmission rate F_{rate} (pulses/s) for this system.

- 2.6-7** Sketch an impulse response $h(t)$ of a non-causal LP system that has an approximate cutoff frequency of 5 kHz. Since many solutions are possible, be sure to properly justify your answer.

- 2.6-8** Two LTIC transmission channels are available: the first has impulse response $h_1(t) = u(t) - u(t-1)$ and the second has impulse response $h_2(t) = \delta(t) + 0.5\delta(t-1) + 0.25\delta(t-2)$. Explain which channel is better suited for the transmission of high-speed digital data (pulses).

- 2.6-9** Consider a linear time-invariant system with impulse response $h(t)$ shown in Fig. P2.6-9. Outside the interval shown, $h(t) = 0$.

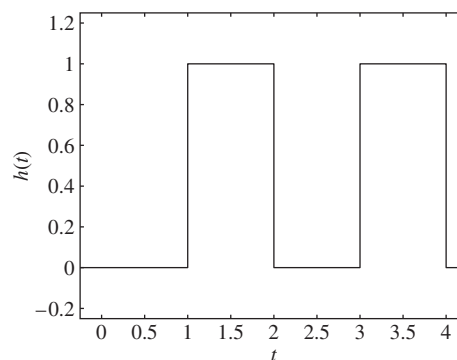


Figure P2.6-9

- (a) What is the rise time T_r of this system? Remember, rise time is the time between the application of a unit step and the moment at which the system has “fully” responded.
 (b) Suppose $h(t)$ represents the response of a communication channel. What conditions might cause the channel to have such an impulse response? What is the maximum

average number of pulses per unit time that can be transmitted without causing interference? Justify your answer.

- (c) Determine the system output $y(t) = x(t) * h(t)$ for $x(t) = [u(t-2) - u(t)]$. Accurately sketch $y(t)$ over $(0 \leq t \leq 10)$.

2.6-10 A lowpass LTIC system has impulse response $h(t) = -te^{-t}u(t)$.

- (a) Accurately sketch $h(t)$.
 (b) Describe a rectangular impulse response $\hat{h}(t)$ as an appropriate approximation of $h(t)$. What is the approximate cutoff frequency of this system?

2.6-11 A lowpass LTIC system has impulse response $h(t)$, as shown in Fig. P2.4-8.

- (a) As discussed in Sec. 2.6-2, determine a rectangular approximation $\hat{h}(t)$ to $h(t)$.
 (b) Using $\hat{h}(t)$, what is the time constant T_h of this lowpass system?
 (c) Using $\hat{h}(t)$, what is the approximate radian cutoff frequency ω_c of this lowpass system?
 (d) Assuming a frequency $\omega_0 \ll \omega_c$, what is the system response $y(t)$ to the input $x(t) = \sin(\omega_0 t + \pi/3)$?

2.6-12 A first CT lowpass system with time constant $T_1 = 4 \mu\text{s}$ is put in series with a second CT lowpass system with time constant $T_2 = 2 \mu\text{s}$. Make an educated sketch of the overall impulse response function $h_{\text{series}}(t)$. What is the time constant T_{series} of the overall series-connected system?

2.7-1 An LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant coefficient linear differential equation:

$$(D^4 - 16)\{y(t)\} = (D - 2)\{x(t)\}.$$

- (a) What are the 4 characteristic roots of this system ($\lambda_1, \lambda_2, \lambda_3$, and λ_4)? Determine the roots by hand and then verify your answers using MATLAB's `roots` command.
 (b) From Eq. (2.17), computing $h(t)$ requires a signal $\tilde{y}_n(t) = \sum_{k=1}^4 c_k e^{\lambda_k t}$. First, determine a matrix representation of the system of equations needed to solve for the four coefficients c_k . Second, write MATLAB code that computes the length-4 column vector of coefficients c_k .

2.7-2 Define $x(t) = 2u(t + \frac{2}{3}) - 2u(t)$. Further, define the periodic signal $h_1(t)$ as

$$h_1(t) = \begin{cases} t & 0 \leq t < 1 \\ h_1(t+1) & \forall t \end{cases}$$

Lastly, define the aperiodic signal $h_2(t)$ in terms of $h_1(t)$ as

$$h_2(t) = h_1(t)[u(t-1) - u(t-2)]$$

- (a) Use MATLAB to plot $x(t)$, $h_1(t)$, and $h_2(t)$ over the interval $-2.5 \leq t \leq 3.5$.
 (b) Using the graphical convolution procedure, compute $y_2(t) = x(t) * h_2(t)$.
 (c) Compute by hand and then MATLAB plot $y_1(t) = x(t) * h_1(t)$. Modify program CH2MP4.m in Sec. 2.7-4 to validate your analytical result.

2.7-3 Consider the circuit shown in Fig. P2.7-3. Assume ideal op-amp behavior and recall that

$$i_C(t) = C \frac{dV_C(t)}{dt}$$

Without a feedback resistor R_f , the circuit functions as an integrator and is unstable, particularly at dc. A feedback resistor R_f corrects this problem and results in a stable circuit that functions as a “lossy” integrator.

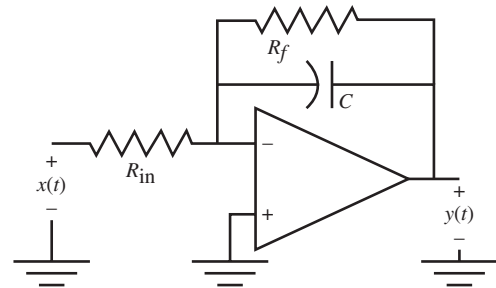


Figure P2.7-3

- (a) Determine the differential equation that relates the input $x(t)$ to the output $y(t)$. What is the corresponding characteristic equation?
 (b) To demonstrate that this “lossy” integrator is well behaved at dc, determine the zero-state

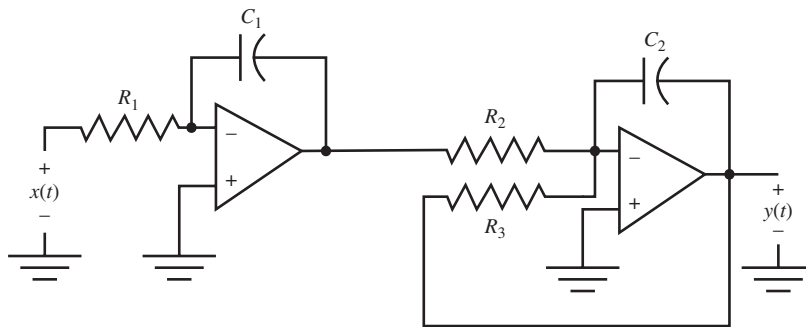


Figure P2.7-4

- response $y(t)$ given a unit step input $x(t) = u(t)$.
- (c) Investigate the effect of 10% resistor and 25% capacitor tolerances on the system's characteristic root(s).
- 2.7-4** Consider the electric circuit shown in Fig. P2.7-4. Let $C_1 = C_2 = 10 \mu\text{F}$, $R_1 = R_2 = 100 \text{ k}\Omega$, and $R_3 = 50 \text{ k}\Omega$.
- (a) Determine the corresponding differential equation describing this circuit. Is the circuit BIBO-stable?
- (b) Determine the zero-input response $y_0(t)$ if the output of each op amp initially reads one volt.
- (c) Determine the zero-state response $y(t)$ to a step input $x(t) = u(t)$.
- (d) Investigate the effect of 10% resistor and 25% capacitor tolerances on the system's characteristic roots.
- 2.7-5** Input $x(t) = 3[u(t) - u(t-1)] + 2[u(t-2) - u(t-3)]$ is applied to a lowpass LTIC system with impulse response $h(t) = (4-t)[u(t) - u(t-2)]$ to produce output $y(t) = x(t) * h(t)$. Modify program CH2MP4.m in Sec. 2.7-4 to perform the graphical convolution procedure to produce a plot of $y(t)$.